Seed Germinator



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Seed germination chamber / incubator

What it is used for

A seed germinator provides controlled temperature, moisture and sometimes light to support seed germination. It helps students and growers observe germination rates and prepare healthy seedlings under suitable conditions.

Main parts

- Identify the insulated chamber and access door.
- Locate the seed trays, racks or germination containers.
- Identify the temperature sensor and display.
- Locate the heating and cooling system, depending on model.
- Identify the humidity control, water tray or misting arrangement, if fitted.
- Locate the lighting controls and lamps, where provided.
- Identify the control panel, power switch and alarm indicators.

Before you start

1. Read the operating instructions and select conditions suitable for the seed species.
2. Inspect the chamber, door seals, trays, sensors and electrical connections for damage.
3. Clean trays and containers using appropriate hygiene procedures.
4. Use viable seeds from a labelled and suitable source.
5. Prepare clean growing medium or germination paper as required.
6. Check that water reservoirs and drainage arrangements are clean and functional.
7. Place the germinator in a stable, ventilated area away from direct sunlight.

How to use

1. Arrange seeds in labelled trays or containers using the recommended method.
2. Add the required moisture without waterlogging the seeds or growing medium.
3. Place the trays evenly inside the chamber without blocking airflow.
4. Close the door and set the specified temperature and humidity conditions.
5. Set the lighting schedule if the selected seeds require light control.
6. Switch on the germinator and allow conditions to stabilise.
7. Monitor temperature, moisture and visible germination at suitable intervals.
8. Add water or adjust settings only as recommended for the seed species.
9. Record germination counts, dates and observations for the learning activity.
10. Remove seedlings carefully when ready and switch off the unit before cleaning.

⚠ Safety

- Keep water away from electrical outlets, controls and damaged cables.
- Do not exceed the manufacturer's temperature or humidity limits.
- Avoid opening the chamber unnecessarily, as this can disturb conditions.
- Use clean hands or suitable gloves when handling seeds and growing materials.
- Disconnect power before cleaning, servicing or accessing internal components.
- Report unusual odours, overheating or faulty sensors to the teacher.

Care & storage

- Clean and dry trays and the chamber after each germination cycle.
- Empty and clean water reservoirs according to the manufacturer's instructions.
- Check sensors, door seals and temperature controls regularly.
- Store the unit dry and protect it from dust and pests.

Common mistakes to avoid

- Setting unsuitable temperature or moisture levels for the seed species.
- Waterlogging seeds and encouraging mould growth or seed rot.
- Overloading trays and blocking airflow inside the chamber.
- Failing to label seed samples and record germination observations.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar